

DRAFT VULNERABILITY ASSESSMENT REPORT

**Climate Change Impact and Vulnerability Assessment for Extreme Climate Events in
Uttarakhand**

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Executive Summary

Uttarakhand's mountainous terrain, fragile geology, and increasing exposure to climate-induced hazards place the state among the most vulnerable regions in the Indian Himalayan Region (IHR). This report presents the Draft Vulnerability Assessment for Uttarakhand, prepared under the technical collaboration between the State Environment Conservation and Climate Change Directorate (SECCCD), Government of Uttarakhand, and GIZ/CAFRI II, building on the Climate Hazard Profile developed by IIT Roorkee. The assessment integrates physical, social, economic, environmental, and institutional dimensions of vulnerability across all 13 districts and their constituent blocks, drawing on the IIT Roorkee climate hazard profile, ETCCDI-based climate indices, and extensive secondary data from line departments.

The state continues to experience a high frequency of climate-induced hazards, including flash floods, landslides, glacial lake outburst floods (GLOFs), cloudbursts, and forest fires, with more than half of Uttarakhand's geographical area classified as highly or very highly susceptible to landslides. Regional climate projections suggest a temperature rise of 1.3-1.6°C by mid-century and 2.3-5.9°C by the century's end under high-emission scenarios, accompanied by a decline in pre-monsoon rainfall and a 6-16% increase in monsoon precipitation, both of which are expected to intensify the frequency and severity of extreme events.

This report develops a multi-tiered Composite Climate Vulnerability Index (CVI) using a sector-disaggregated framework that classifies indicators into the three conventional IPCC vulnerability components , exposure, sensitivity, and adaptive capacity , supplemented by environmental and institutional dimensions specific to the Himalayan context.

Sector-specific vulnerability indices were developed for agriculture and livelihoods, water resources, human health, roads and critical infrastructure, tourism, and energy. These reveal that

The risk assessment component combines hazard probabilities derived from the IIT Roorkee climate hazard profile with the composite vulnerability scores to generate block-level risk profiles and a risk matrix that differentiates blocks by likelihood and consequence. This analysis identifies a set of high-priority blocks, concentrated along the

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The report concludes with a prioritised set of adaptation and mitigation measures organised by sector, ranging from nature-based solutions such as spring-shed rejuvenation and slope stabilisation through bioengineering, to infrastructure measures including climate-resilient road design and early warning systems, and social protection measures such as livelihood diversification and targeted support for women-headed households and marginalised communities. A comprehensive stakeholder map identifies the institutional roles and coordination mechanisms required across state-line departments to operationalise these recommendations.

1. Introduction

Purpose of the Vulnerability & Risk Assessment

Uttarakhand's exposure to recurrent natural hazards , flash floods, landslides, earthquakes, forest fires, GLOFs, and cloudbursts , has historically disrupted development gains across the state, with major events such as the 2013 Kedarnath floods, the 2016 forest fires, and the 2021 Chamoli rock-ice avalanche each causing significant loss of life, infrastructure, and livelihoods. As regional climate projections point to rising temperatures, increasing monsoon variability, and accelerated glacier melt, the probability and intensity of such extreme events is expected to grow, with landslide risk in certain districts projected to rise by as much as 90% during the monsoon months by the latter half of the century.

This Vulnerability Assessment has been commissioned to systematically evaluate the vulnerability of people, infrastructure, public utilities, and economic activities across Uttarakhand to these climate-induced hazards, under both current conditions and future growth and climate scenarios extending to 2080. The assessment serves three interlinked purposes:

- To quantify the relative vulnerability of each of Uttarakhand's 13 districts and their constituent blocks using a structured, indicator-based methodology that is transparent, replicable, and updatable over time.
- To identify the causal drivers of vulnerability , physical, social, economic, environmental, and institutional , so that adaptation investments can be targeted at root causes rather than symptoms.
- To combine vulnerability outcomes with hazard probability data to generate a block-wise risk profile that can directly inform the prioritisation of adaptation and mitigation measures within state, district, and city-level development plans.

The assessment is designed to feed directly into the Digital Vulnerability Atlas being developed under this assignment, providing the analytical and statistical backbone for the spatial layers that the Atlas will visualise.

Linkage with Sectoral Climate Impact Assessment

This Vulnerability Assessment is one of two analytically linked components of the broader climate change impact study being undertaken for SECCCD under GIZ/CAFRI II. The companion Sectoral Climate Impact Assessment uses IIT Roorkee's downscaled climate projections under SSP2-4.5 and SSP5-8.5 scenarios to evaluate impacts on eight priority sectors , tourism, forestry, agriculture and livelihoods, water resources, roads and critical infrastructure, energy, health, and water supply and sanitation , for the mid-century (2021-2050) and end-century (2071-2100) horizons.

The two assessments are deliberately structured to be complementary rather than duplicative. The Sectoral Climate Impact Assessment establishes what is likely to change in each sector under future climate scenarios , for instance, projected shifts in crop suitability, water availability, or heat stress days. This Vulnerability Assessment instead establishes who and where is most exposed and least able to cope with those changes, using current socio-economic, infrastructural, and institutional indicators as the basis

for a composite vulnerability score. The sectoral impact layers generated by the companion assessment are used in this report as one of several inputs into the exposure and sensitivity dimensions of the vulnerability framework, particularly for agriculture, water resources, and health.

Both assessments draw on a single, shared scientific foundation , the IIT Roorkee Climate Hazard Profile, developed using IMDAA, CHIRPS, CHIRTS, and downscaled CMIP6 datasets, together with ETCCDI-defined climate indices , ensuring that hazard and impact narratives remain internally consistent across both reports and across the eventual Digital Vulnerability Atlas.

Report Structure

This report is organised into eight sections. Section 2 sets out the conceptual framework for climate vulnerability used in this assessment, situates Uttarakhand's existing vulnerability status across five dimensions, and defines the three core IPCC-aligned components of vulnerability , exposure, sensitivity, and adaptive capacity , that structure the indicator framework. Section 3 describes the development of the Vulnerability Index, including the indicator selection process, data sources, normalisation and weighting methodology, and presents illustrative block-wise results.

Section 4 presents sector-specific vulnerability indices for agriculture and livelihoods, water resources, human health, roads and infrastructure, tourism, and energy. Section 5 provides a comprehensive stakeholder map identifying the departments and institutions relevant to vulnerability reduction and their interdependencies. Section 6 aggregates the sectoral indices into a Composite Climate Vulnerability Index (CVI), presents district-level vulnerability classification and hotspot analysis, and projects future vulnerability shifts under SSP climate scenarios.

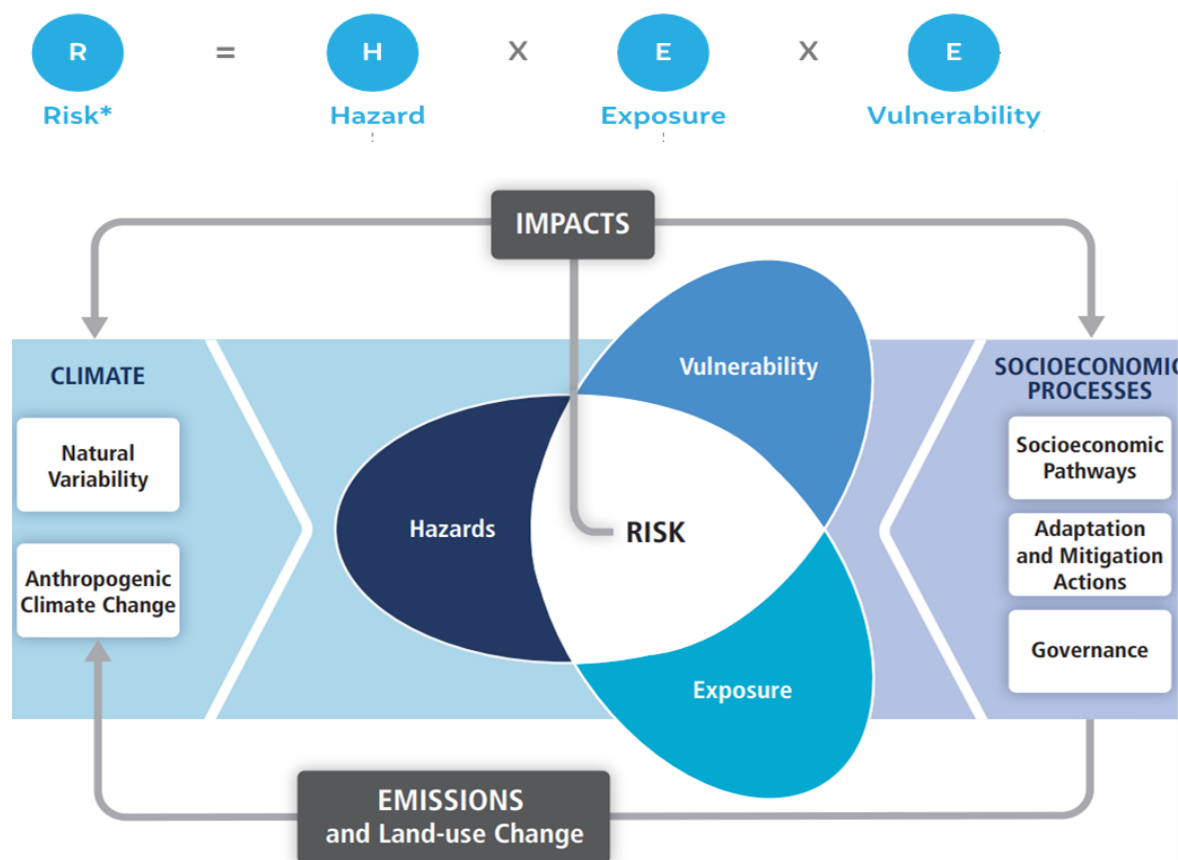
Section 7 combines hazard probabilities with the vulnerability indices to generate block-level risk profiles and a risk matrix identifying high-risk areas. Section 8 concludes with a prioritised set of adaptation and mitigation measures organised by sector, including nature-based solutions, infrastructure investments, and social protection measures, linked explicitly to the risk hotspots identified in Section 7.

2. Vulnerability Assessment Framework

Conceptual Framework for Climate Vulnerability

This assessment is structured in full alignment with the conceptual architecture set out by the Intergovernmental Panel on Climate Change (IPCC), integrating both the AR5 vulnerability framing and the AR6 risk framing so that the report's outputs are consistent with current international practice while remaining fit for the sector-disaggregated, block-level analysis required under this assignment.

Under the IPCC Fifth Assessment Report (AR5) framing, adopted by Working Group II, vulnerability is defined as a function of exposure, sensitivity, and adaptive capacity. Exposure refers to the presence of people, livelihoods, species, ecosystems, environmental functions and resources, infrastructure, or economic, social, and cultural assets in places and settings that could be adversely affected by climate hazards. Sensitivity refers to the degree to which a system is affected, either adversely or beneficially, by climate-related stimuli. Adaptive capacity refers to the ability of a system to adjust to climate change, moderate potential damages, take advantage of opportunities, or cope with consequences. This three-component structure is retained in this report as the operative basis for indicator classification and index computation, since it is the formulation most directly suited to constructing a quantitative, indicator-based composite vulnerability index of the kind required under this assignment.



The IPCC Sixth Assessment Report (AR6) subsequently embedded this same vulnerability concept within a broader risk framework, in which climate risk arises from the dynamic interaction between hazard, exposure, and vulnerability, with vulnerability itself disaggregated into sensitivity and the lack of capacity to cope and adapt. Under AR6, hazard refers to the potential occurrence of a natural or human-induced physical event or trend that may cause loss of life, injury, or other health impacts, as well as damage and loss to property, infrastructure, livelihoods, service provision, ecosystems, and environmental resources; exposure and vulnerability jointly determine the degree to which a given hazard translates into realised risk. AR6 also foregrounds the role of human responses, adaptation and mitigation actions, as dynamic drivers that continuously reshape both vulnerability and exposure over time, rather than treating them as static background conditions.

This report applies the AR5 exposure-sensitivity-adaptive capacity structure in Sections 3 and 4, where the task is to construct sector-specific and composite vulnerability indices from indicator data, since this is the formulation that is operationally tractable for indicator-based quantification at block level. The report then applies the full AR6 risk framework in Section 7, where vulnerability outputs from Sections 3, 4, and 6 are explicitly combined with independently derived hazard probabilities, drawn from the IIT Roorkee Climate Hazard Profile, to compute block-level climate risk, consistent with the AR6 formulation of risk as arising from the interaction of hazard, exposure, and vulnerability. This two-stage structure, in which AR5-consistent vulnerability index construction feeds into an AR6-consistent hazard-exposure-vulnerability risk computation, ensures the assessment is methodologically aligned with both IPCC formulations rather than conflating vulnerability and risk, which the IPCC AR6 glossary explicitly distinguishes as related but non-equivalent concepts.

This framework has been further adapted to the specific geophysical, socio-economic, and institutional context of the Indian Himalayan Region (IHR) by reviewing existing vulnerability assessment studies and frameworks applied within the region, including state action plans on climate change, NITI Aayog's Himalayan State Working Group reports, India's national Climate Vulnerability Assessment for the Indian Himalayan Region (DST, 2018-19), and academic vulnerability studies conducted for comparable Himalayan geographies such as Himachal Pradesh and Sikkim. This review informed the selection of indicators that are analytically sound, measurable at the block level using available secondary data, spatially comprehensive across all 13 districts, and meaningfully related to the climate hazards prevalent in Uttarakhand, while remaining traceable to the IPCC AR5 and AR6 definitions adopted above.

The framework recognises that vulnerability in a mountain state cannot be captured through the conventional exposure-sensitivity-adaptive capacity triad alone without further contextualisation. Terrain fragility, ecosystem degradation, and the strength of local governance and institutional response mechanisms play an outsized role in determining how a hazard event translates into loss and damage, and these factors fall well within the scope of what AR6 treats as determinants of sensitivity and adaptive capacity. Accordingly, this assessment organises indicators relating to these factors under two cross-cutting dimensions, environmental and institutional, detailed in Section 2.b below, which are explicitly mapped onto the IPCC exposure-sensitivity-adaptive capacity structure for index computation in Section 3, with environmental indicators predominantly informing exposure and sensitivity, and institutional indicators predominantly informing adaptive capacity. No indicator is treated as falling outside the IPCC

three-component structure; the environmental and institutional dimensions serve as an organisational lens for indicator selection rather than a fourth and fifth component running parallel to the IPCC framework.

Uttarakhand Existing Vulnerability Status

Uttarakhand's vulnerability to climate-induced hazards is shaped by the interaction of five overlapping dimensions: physical, social, economic, environmental, and institutional. Each is summarised below based on the existing evidence base, prior to the indicator-level quantification presented in Section 3.

Physical Dimension

The physical dimension captures the degree to which settlements, infrastructure, and the built environment are exposed to and fragile against climate hazards. More than half of Uttarakhand's geographical area is classified as highly or very highly susceptible to landslides, with steep slopes, fragile geology, and high drainage density compounding the risk from cloudburst-triggered debris flows. Settlements located along river corridors, particularly along the Bhagirathi, Alaknanda, Mandakini, and Kali river systems, face recurring flash flood exposure, while several mid- and high-elevation habitations remain dependent on single-lane hill roads that are frequently disrupted by landslides during the monsoon. Critical infrastructure such as hydropower installations, bridges, and health facilities sited in valley floors face elevated exposure to both fluvial flooding and GLOF-related debris flows, while remoteness and difficult terrain access continue to constrain the speed of post-disaster response in several blocks of Pithoragarh, Chamoli, and Uttarkashi.

Social Dimension

The social dimension reflects the demographic composition and social vulnerability of Uttarakhand's population. The state's hill districts have experienced sustained out-migration of working-age males in search of livelihood opportunities, resulting in a demographic profile increasingly composed of women, children, and the elderly in many hill blocks, a pattern with direct implications for disaster preparedness and response capacity at the household level. Access to basic services, including healthcare, education, and emergency communication, declines markedly with elevation and remoteness, while literacy and awareness of climate risk vary considerably across districts. Marginalised groups, including Scheduled Castes, Scheduled Tribes (notably the Van Gujjars and Bhotia communities), single-parent and women-headed households, and persons with disabilities, face compounded barriers to evacuation, relief access, and recovery support during and after extreme events.

Economic Dimension

Uttarakhand's hill economy remains heavily dependent on climate-sensitive sectors, particularly rain-fed subsistence agriculture, horticulture, animal husbandry, and tourism, with limited income diversification at the household level in many hill blocks. Landholdings are highly fragmented, asset bases are modest, and access to formal credit and crop or livelihood insurance remains limited outside the major urban and plains districts. This narrow economic base reduces households' capacity to absorb shocks from crop loss,

road closures that disrupt tourism inflows, or damage to livestock and horticultural assets, and constrains post-disaster recovery without external support.

Environmental Dimension

Environmental degradation acts as both a driver and an amplifier of climate vulnerability in Uttarakhand. Deforestation and forest fires, 11,256 forest fire incidents were recorded between November 2023 and June 2024 alone across 11 of the state's 13 districts, degrade slope-stabilising vegetation cover and increase runoff and erosion risk in subsequent monsoon seasons. Glacial retreat, with the state's approximately 1,950 glaciers having lost around 16 gigatons of ice between 2000 and 2014, is altering streamflow seasonality and increasing the formation and expansion of glacial lakes prone to outburst flooding. Land degradation and unsustainable groundwater and spring-water extraction are reducing the resilience of natural water systems on which both agriculture and domestic water supply depend, particularly in the mid-hill belt where spring discharge has been widely reported to be declining.

Institutional Dimension

The institutional dimension captures the strength and coordination of governance systems responsible for disaster risk reduction and climate adaptation. Uttarakhand has established a State Disaster Management Authority and district-level disaster management structures, alongside the State Environment Conservation and Climate Change Directorate's mandate for climate action planning. However, capacity for hazard monitoring, early warning dissemination, and inter-departmental coordination varies considerably across districts and blocks, and is generally weaker in remote, high-elevation administrative units that are simultaneously the most physically exposed. Section 5 of this report develops a detailed stakeholder map identifying the specific departmental roles and coordination gaps relevant to closing this institutional vulnerability gap.

Components of Vulnerability

Building on the conceptual framework above, this assessment structures all indicators- including those drawn from the physical, social, economic, environmental, and institutional dimensions described above into the three core IPCC AR5 components for the purpose of index construction: exposure, sensitivity, and adaptive capacity. These three components, in turn, populate the vulnerability term of the IPCC AR6 risk equation applied in Section 7, where vulnerability (sensitivity and the inverse of adaptive capacity) is combined with independently assessed exposure and hazard probability to compute climate risk.

Exposure

Consistent with the IPCC AR5/AR6 definition, exposure indicators capture the degree to which people, assets, and economic activities in a given block are physically present in areas subject to climate hazards, for example, the proportion of settlements within mapped flood or landslide hazard zones, road length traversing high-susceptibility terrain, or area under climate-sensitive crops in drought- or flood-prone tracts. In the AR6 risk framework, exposure is treated as conceptually distinct from vulnerability, though for the purposes of constructing the composite indices in Sections 3, 4, and 6, exposure indicators are

computed alongside sensitivity and adaptive capacity indicators and only disaggregated again for the explicit hazard-exposure-vulnerability risk computation in Section 7.

Sensitivity

Sensitivity indicators, the first of the two IPCC AR6 sub-components of vulnerability, capture the degree to which a given population or sector is likely to be adversely affected once exposed to a hazard , for example, dependence on rainfed agriculture, proportion of marginal and small landholdings, prevalence of climate-sensitive disease vectors, or reliance on single-source water supply.

Adaptive Capacity

Adaptive capacity indicators, the second IPCC AR6 sub-component of vulnerability (expressed as the inverse of coping and adaptive capacity), capture the resources and mechanisms available to anticipate, cope with, and recover from hazard impacts , for example, access to formal credit and insurance, density of health infrastructure, presence of early warning systems, road network redundancy, and institutional capacity for disaster response. Adaptive capacity indicators are normalised in the negative direction, since higher adaptive capacity reduces overall vulnerability, while exposure and sensitivity indicators are normalised in the positive direction.

3. Vulnerability Index Development

Framework

The Vulnerability Index for Uttarakhand is constructed as a composite, multi-sector indicator framework following the IPCC AR5 Exposure-Sensitivity-Adaptive Capacity (E-S-AC) structure adopted in Section 2, computed independently for each sector and then aggregated into sectoral and composite indices. The framework is designed to operate at the block level, the finest administrative unit for which the relevant secondary data are consistently available across all 13 districts, with results subsequently aggregable to the district level for comparison with state averages and for the cluster classification presented in Section 6. The composite and sectoral vulnerability indices developed in this section and in Section 4 correspond to the vulnerability term of the IPCC AR6 risk equation; they are deliberately kept independent of hazard probability at this stage, with the combination of vulnerability and hazard into climate risk performed explicitly in Section 7.

Eight sectors are covered within the framework: roads and critical infrastructure, agriculture and livelihoods, water resources, tourism, human health, energy, and the cross-cutting social and economic dimensions described in Section 2.b. Each sector-specific index is built from a curated indicator basket, with each indicator assigned to one of the three IPCC components (exposure, sensitivity, or adaptive capacity) and a normalisation direction reflecting whether higher values of the raw indicator increase or decrease vulnerability.

Methodology

The methodology follows five sequential steps, consistent with established practice in composite vulnerability indices such as those used in India's Climate Vulnerability Assessment for the Indian Himalayan Region (DST, 2018-19) and subsequent state-level adaptations:

1. Indicator selection and assignment to exposure, sensitivity, or adaptive capacity components, with each indicator's data source identified and its normalisation direction specified (Section 3.c).
2. Data compilation at the block level from departmental and census sources, with gap-filling through district-level disaggregation where block-level data are unavailable.
3. Normalisation of each indicator to a common 0-1 scale using min-max normalisation, applied in the direction specified for that indicator (Section 3.d).
4. Weighting and aggregation of normalised indicators within each component, and of the three components into a sector-specific vulnerability index, using the equal-weight and Principal Component Analysis (PCA) approaches described in Section 3.d.
5. Classification of resulting block- and district-level scores into six vulnerability categories, very low, low, moderate, high, very high, and extremely high, using quantile-based or natural-breaks (Jenks) classification, consistent with the categorisation applied in Section 6.

This sequence is applied independently for each of the sector-specific indices presented in Section 4, and the resulting sectoral indices are subsequently combined into the Composite Climate Vulnerability Index (CVI) described in Section 6.

Indicator Selection and Justification

Indicators were selected against four criteria specified in the Terms of Reference: analytical soundness, measurability at the block level using available data sources, spatial coverage across all 13 districts, and relevance to the climate hazards prevalent in Uttarakhand. The selection draws on a review of vulnerability assessment frameworks previously applied in the Indian Himalayan Region, supplemented by indicators specific to Uttarakhand's hazard profile (landslide, GLOF, and cloudburst exposure in particular).

The table below sets out the indicator basket for the roads, agriculture, and water resources sectors as illustrative examples of the indicator selection logic:

Indicator	Meaning	Dimension	Category	Rationale
Annual Rainfall	From the baseline data	Hazard	Environmental	Determines the overall availability of renewable water resources. Lower rainfall reduces recharge and streamflow.
Drought Frequency	Number of drought years in last 20–30 years	Hazard	Environmental	Frequent meteorological droughts reduce surface water availability and trigger spring discharge decline.
Extreme Rainfall Frequency	Avg. days/year with rainfall ≥ 64.5 mm	Hazard	Environmental	Cloudbursts damage pipelines, springs and storage structures while contaminating drinking water sources.

Mean Maximum Temperature	Annual average of daily maximum temperature (°C)	Hazard	Environmental	Higher temperatures increase evapotranspiration, reducing effective water availability.
Change in Maximum Number of Consecutive Dry Days (CDD)	Maximum annual consecutive dry days	Hazard	Environmental	Longer dry spells reduce recharge and increase seasonal water scarcity.
Flood Hazard	% flood-prone	Hazard	Environmental	Floods damage drinking water infrastructure and contaminate water sources.
Landslide Hazard	% landslide-prone villages OR % block area in landslide hazard zone	Hazard	Environmental	Landslides destroy springs, pipelines and water supply systems.
Population Density	Population/km ²	Physical	Exposure	Higher population increases domestic water demand and exposure to shortages.
Water Infrastructure Density	Gools, canals, ponds, tanks and check dams per 100 km ²	Physical	Exposure	More water infrastructure means greater assets exposed to climate hazards.

Domestic Water Demand	Population × LPCD × 365	Physical	Exposure	Greater demand increases pressure on available water resources.
Irrigation Intensity	Gross Irrigated Area / Net Irrigated Area × 100	Physical	Exposure	Higher irrigation demand increases dependence on reliable water supplies.
Watershed layers		Physical	Exposure	High abstraction increases pressure on watersupply
Tribal Population Density	ST Population/km ²	Social	Sensitivity	Remote tribal communities rely heavily on springs and local water resources.
Rainfall Variability (CV)	Coefficient of Variation of annual rainfall	Environmental	Sensitivity	High variability reduces reliability of water resources even where annual rainfall is high.
Groundwater Level	Average pre-monsoon groundwater depth (mbgl)	Environmental	Sensitivity	Declining groundwater reduces water availability during dry periods.

Spring Density	Number of perennial springs/km ²	Environmental	Sensitivity	Blocks with few springs are more vulnerable to water scarcity.
Settlements Lacking Drinking Water Facility	settlement not covered- without adequate drinking water supply	Physical	Sensitivity	Indicates existing water insecurity.
Volumetric Soil Moisture Content	Average soil moisture (%)	Environmental	Sensitivity	Low soil moisture limits recharge and increases drought sensitivity.
Evapotranspiration (ET)	Mean ET (mm/day)	Environmental	Sensitivity	Higher ET reduces effective water availability.
Drainage Density	Total stream length / Area (km/km ²)	Environmental	Sensitivity	Controls runoff, infiltration and groundwater recharge potential.
Tap Water Supply Coverage (FHTC)	Functional Household Tap Connections / Total Households ×100	Physical	Adaptive Capacity	Reliable household water supply improves resilience to climate variability.
Treated Watersheds	% watershed area treated	Institutional	Adaptive Capacity	Watershed treatment improves recharge, reduces runoff and sustains springs.

Canal Irrigation (%)	Canal Irrigated Area / Total Net Irrigated Area $\times 100$	Physical	Adaptive Capacity	Diversifies irrigation sources and reduces dependence on rainfall.
Forest Cover	Forest Area / Geographical Area $\times 100$	Environmental	Adaptive Capacity	Forests enhance infiltration, recharge and watershed protection.
Energised Private Tubewells/Pump Sets	Number per 1000 ha irrigated area	Physical	Adaptive Capacity	Provides alternative irrigation during water shortages, particularly in plains.
Net Sown Area (%)	Net Sown Area / Reported Area $\times 100$	Physical	Exposure	Larger cultivated area increases agricultural assets exposed to climate hazards.
Foodgrain Dependence	Foodgrain / Gross Sown	Economic	Exposure	Monocropping and foodgrain dependence increase exposure to climate variability.
Irrigation Infrastructure	Canal length, Gools, Hauz, Haidram	Physical	Exposure	Climate hazards affecting irrigation infrastructure reduce agricultural production.
% population dependent on agriculture	(Farmers + Agricultural Labourers)/Population $\times 100$	Social	Exposure	Households whose primary occupation is cultivation may have better resource access

				(land ownership) but face elevated income risk if crop failures occur. Greater involvement in farming (> 6 months) signifies higher sensitivity due to reliance on potentially vulnerable agricultural systems
Current Fallow land (%)	Current Fallow / Reported Area ×100	Environmental	Exposure	Indicates agricultural land temporarily left uncultivated due to climatic or socioeconomic stress.
Annual Rainfall		Environmental	Exposure	
NDVI		Environmental	Exposure	
Livestock density		Social	Exposure	Higher livestock concentration increases exposure of livestock-based livelihoods.
Horticulture Area (% of Gross Sown Area)			Exposure	Rice is highly climate sensitive
Agricultural Labourers (% of workers)	Agricultural Labourers / Total Workers ×100	Social	Sensitivity	Agricultural labourers depend on seasonal farm employment and are highly

				vulnerable to crop failures.
Volumetric Soil Moisture content (%)	Average growing-season soil moisture	Environmental	Sensitivity	Lower soil moisture directly increases crop water stress and drought sensitivity.
% of marginal and small landholdings	Number of marginal and small operational holdings, i.e., up to 2 hectares/Total number of operational holdings	Economic	Sensitivity	Marginal and smallholder farmers experience immediate hardship in face of any climatic hazard. They are unable to make adequate decisions about when to sow, what to grow, and how-to time inputs. (Sathyan, et. al., 2018). They also find it difficult to cope with the high food price and the fluctuations in the same.
Average Operational Holding Size	Total cultivated area / Total holdings (Ha) — invert before index	Economic	Sensitivity	Smaller holdings reduce economies of scale and increase vulnerability.
Degraded area	Porportion of degraded area to block area	Environmental	Sensitivity	

Area under rainfed agriculture	(Net sown area - Net irrigated area)/ Net sown area	Environmental	Sensitivity	Rainfed agriculture is highly sensitive to the vagaries of weather. Lack of irrigation indicates a lack of adaptive capacity to mitigate the impacts of climate risks, leading to increased crop loss and reduced income of households dependent on rainfed agriculture (Rani, et. al., 2011)
% of ST/SC population	SC + ST Population / Total Population ×100	Social	Sensitivity	Scheduled Tribes (ST) and Scheduled Castes (SC) often face socioeconomic marginalization and rely heavily on natural resources. A larger ST/SC population typically indicates higher sensitivity to resource scarcity, as these communities may have limited adaptive capacity and

				fewer livelihood options
% of net irrigated area in (ha)	Net Irrigated Area / Net Sown Area ×100	Environmental	Sensitivity	Irrigation reduces dependence on rainfall and stabilizes production.
Cropping Intensity	Gross Sown / Net Sown × 100	Physical	Sensitivity	Higher cropping intensity increases dependence on stable climate and water availability.
Literacy rate (% of literate population)	Literate Population / Total Population ×100	Social	Adaptive Capacity	Higher literacy improves access to information, technology and adaptation options.
Agriculture credit	% village access to agricultural credit societies	Institutional	Adaptive Capacity	Institutional credit improves farmers' capacity to invest in adaptation.
Availability of KSKs	Krishi Seva Kendras /10,000 farmers	Institutional	Adaptive Capacity	Extension services promote climate-resilient technologies and practices.
% villages connected only by seasonal roads	Seasonal road villages / Total villages ×100 (inverse)	Physical	Adaptive Capacity	Poor connectivity limits market access, extension services and emergency support.

Cold Storage Capacity	MT /1000 ha cropped area	Physical	Adaptive Capacity	Reduces post-harvest losses and enhances resilience of perishable crops.
Distance to fair market shop	in km (inverse)	Physical	Adaptive Capacity	Poor connectivity limits market access, extension services and emergency support.
Farmer Beneficiary	Beneficiary Farmers / Total Farmers ×100	Institutional	Adaptive Capacity	Government support programmes strengthen farmers' adaptive capacity.
Fertilizer consumption	kg/ha gross sown area	Economic	Adaptive Capacity	Reflects access to agricultural inputs that can improve productivity when used appropriately.
Seed Godowns/Fertiliser Depot	Number per 10,000 farmers	physical	Adaptive Capacity	Ensures timely access to quality seeds and agricultural inputs.
Health Facility Density	PHCs + CHCs + Hospitals per 100 km ²	Physical	Exposure	More health infrastructure is exposed to floods, landslides and extreme weather.
Population Served by Health Facilities	Population per PHC/CHC	Social	Sensitivity	Higher population burden reduces

				healthcare accessibility during climate emergencies.
Villages >5 km from Health Facility	% villages located >5 km from nearest health facility	Social	Sensitivity	Remote communities have reduced access during disasters.
Institutional Deliveries (%)	Institutional deliveries / Total deliveries ×100	Social	Adaptive Capacity	Indicates accessibility and resilience of healthcare services.
Doctors Availability	Doctors per 10,000 population	Institutional	Adaptive Capacity	Greater medical capacity improves emergency response.
Ambulance Density	Ambulances per 100,000 population	Physical	Adaptive Capacity	Faster emergency response during disasters.
Health Facilities with Functional Water Supply	% health facilities with piped water	Physical	Adaptive Capacity	Ensures continuity of healthcare during climate events.
Health Facilities with Backup Power	% health facilities with DG/Solar backup	Physical	Adaptive Capacity	Maintains healthcare services during power outages.
Road Density	Total road length (km)/100 km ²	Physical	Exposure	Larger road network increases infrastructure exposed to hazards.
Critical Road Length	NH + SH + MDR length (km)	Physical	Exposure	Failure of critical transport corridors causes cascading impacts.

Bridges Density	Number of bridges/100 km ²	Physical	Exposure	Bridges are highly susceptible to floods and landslides.
Villages Connected by All-weather Roads	% villages connected by all-weather roads	Physical	Adaptive Capacity	Reliable access improves emergency response and recovery.
Seasonal Road Connectivity	% villages connected only by seasonal roads	Physical	Sensitivity	Seasonal roads become inaccessible during monsoon.
Alternate Route Availability	Number of alternate access routes	Physical	Adaptive Capacity	Reduces isolation following road failures.
Road Maintenance Expenditure	₹/km/year	Institutional	Adaptive Capacity	Better maintenance improves resilience.
Average Travel Time to District HQ	Hours	Physical	Sensitivity	Longer travel times reduce emergency accessibility.
Power Infrastructure Density	Substations + Transformers + Transmission Towers per 100 km ²	Physical	Exposure	More infrastructure is exposed to climate hazards.
Transmission Line Density	km of transmission lines/100 km ²	Physical	Exposure	Longer transmission networks have greater exposure to storms and landslides.
Hydropower Capacity	Installed hydropower (MW)	Physical	Exposure	Hydropower assets are exposed to floods, sedimentation and GLOFs.

Population without Electricity	% households without electricity	Social	Sensitivity	Lower access increases vulnerability.
Distribution Losses	AT&C Loss (%)	Institutional	Sensitivity	High losses indicate weaker network resilience.
Renewable Energy Share	Solar + Wind + Small Hydro / Total Installed Capacity (%)	Environmental	Adaptive Capacity	Diversified energy sources improve resilience.
Backup Power Availability	% critical facilities with backup power	Physical	Adaptive Capacity	Ensures continuity during outages.
Underground Power Lines	% underground distribution network	Physical	Adaptive Capacity	Less vulnerable to storms and falling trees.
Smart Grid Coverage	% feeders under smart monitoring	Institutional	Adaptive Capacity	Improves rapid restoration after disruptions.

Table 3.1: Illustrative indicator basket for Roads, Agriculture, and Water Resources sectors. (+) indicates the raw indicator increases vulnerability when higher; (–) indicates it reduces vulnerability when higher (adaptive capacity indicators).

Three additional sectors: tourism, health, and energy are populated using an analogous structure. For tourism, indicators include the share of district GDP/employment derived from tourism, seasonal concentration of visitor inflows, and exposure of tourist infrastructure (hotels, homestays, trekking routes) to hazard zones. For health, indicators include health facility density per capita, distance to the nearest functional Community Health Centre, prevalence of climate-sensitive disease burden (vector-borne and water-borne disease incidence), and availability of emergency referral transport. For energy, indicators include dependence on hydropower exposed to GLOF/flood risk, transmission line exposure to landslide zones, and reliability of grid supply during monsoon disruptions. The full indicator baskets for these three sectors, including data sources and normalisation directions, are detailed in Section 4.

Vulnerability Index , Normalisation, Weighting, Aggregation Method

Normalisation

Each raw indicator is normalised to a common 0-1 scale using min-max normalisation, applied across all blocks for which data are available. For positively-oriented indicators (where a higher raw value increases vulnerability), the normalised value is computed as $(X - X_{\min}) / (X_{\max} - X_{\min})$. For negatively-oriented indicators (adaptive capacity indicators, where a higher raw value reduces vulnerability), the normalised value is computed as $(X_{\max} - X) / (X_{\max} - X_{\min})$, so that in both cases a higher normalised score consistently denotes higher vulnerability.

Weighting

Two weighting approaches are applied in parallel for robustness, consistent with practice in comparable Himalayan vulnerability studies. The primary approach applies equal weights to all indicators within a component, and equal weights across the three components (exposure, sensitivity, adaptive capacity) when aggregating to the sector index. This approach is transparent, avoids subjective weighting judgments, and is straightforward to communicate to non-technical stakeholders and government partners. As a robustness check, Principal Component Analysis (PCA) is applied to derive data-driven weights based on the variance structure of the indicator set; where PCA-derived weights diverge materially from the equal-weight results, this is flagged for stakeholder discussion during the validation phase, since substantial divergence may indicate redundancy or multicollinearity among indicators that warrants review.

Aggregation

Within each component, normalised indicator scores are averaged to produce a component sub-index (Exposure sub-index, Sensitivity sub-index, Adaptive Capacity sub-index). The sector vulnerability index is then computed as:

$$\text{Sector VI} = (\text{Exposure sub-index} + \text{Sensitivity sub-index} + \text{Adaptive Capacity sub-index}) / 3$$

with the Adaptive Capacity sub-index entered using its already-inverted normalised values (such that higher adaptive capacity reduces, rather than increases, the sector index). This aggregation is performed independently at the block level for each sector in Section 4, and the resulting sector indices are subsequently combined into the Composite CVI in Section 6 using the same equal-weight logic across sectors, again cross-checked using PCA-derived sector weights.

Data Sources, Validation Approach, and Methodological Limitations

The indicator framework presented in Sections 3.c and 3.d draws on secondary data from a wide range of state line departments and national data sources, consolidated and cross-checked for internal consistency prior to normalisation.

Three methodological limitations are noted at this draft stage and will be addressed during the validation phase. First, block-level data availability is uneven across departments; certain indicators (for instance, spring discharge trend data) are available only for a subset of hill blocks at present, requiring either district-level disaggregation as an interim measure or targeted primary data collection. Second, the equal-weighting approach adopted as the primary method in Section 3.d, while transparent, does not differentiate the relative importance of indicators within a component; the parallel PCA-based weighting is intended to surface cases where this simplification materially affects results, and a structured expert elicitation exercise with sector specialists is recommended as a further robustness check prior to finalisation. Third, the min-max normalisation method used in Section 3.d is sensitive to outliers in the underlying indicator distributions; outlier treatment (winsorisation at the 5th/95th percentile) will be applied during the validation phase where extreme values are identified as data quality issues rather than genuine extreme conditions.

4. Sector-Specific Vulnerability Indices

This section presents sector-specific vulnerability indices for six priority sectors, each constructed using the IPCC AR5 exposure-sensitivity-adaptive capacity methodology described in Section 3. Each sub-section sets out the indicator basket, illustrative normalised component scores, and the resulting sector vulnerability index for a representative set of districts, together with a brief interpretation of the principal drivers of vulnerability in that sector.

4.1 Agriculture and Livelihoods Vulnerability Index (AGVI)

Agriculture remains the principal livelihood base for the majority of Uttarakhand's hill population, and is correspondingly the sector with the highest sensitivity to climate variability across the indicator framework. The AGVI combines exposure to drought and flood-prone tracts, sensitivity arising from rainfed dependence and landholding fragmentation, and adaptive capacity reflected in access to irrigation, credit, and insurance.

Table 4.1: Illustrative district-level Agriculture Vulnerability Index (AGVI) scores (0–1 scale; higher = more vulnerable).

The highest AGVI scores are concentrated in the upper Himalayan districts, where rainfed dependence exceeds 80% of net sown area in several blocks, landholdings are predominantly marginal, and access to formal crop insurance remains limited. Plains districts show substantially lower sensitivity owing to assured irrigation coverage and more diversified cropping patterns, though localised flood-exposed tracts along the Ganga and Kosi rivers register elevated exposure sub-scores even within these otherwise lower-vulnerability districts.

4.2 Water Resources Vulnerability Index (WRVI)

The Water Resources Vulnerability Index captures the dual stress facing Uttarakhand's hill water systems: declining spring discharge linked to land-use change and erratic rainfall, and rising demand from growing settlements and tourism inflows. The index integrates dependence on single-source supply, spring/source discharge trends, and infrastructure coverage for storage and treatment.

Table 4.2: Illustrative district-level Water Resources Vulnerability Index (WRVI) scores.

Almora and Pithoragarh emerge as the most water-stressed districts in the illustrative results, consistent with widely reported spring discharge decline across the mid-hill belt. Haridwar, despite high water demand from agriculture and industry, scores lower on the composite WRVI owing to assured canal and groundwater-based supply, though this masks more localised stress in specific flood-affected or fluoride-affected blocks that will be visible in the full block-level results.

4.3 Human Health / Climate Extremes Vulnerability Index (CLVI)

The CLVI assesses vulnerability arising from the interaction between climate-sensitive health burdens (heat stress, vector-borne and water-borne disease, and disaster-related injury) and the adequacy of health system infrastructure. Indicators include facility density, distance to the nearest functional health centre, disease incidence trends, and the availability of emergency referral transport.

Table 4.3: Illustrative district-level Health/Climate Extremes Vulnerability Index (CLVI) scores.

Remote, high-elevation districts show the highest CLVI scores, driven primarily by low adaptive capacity , specifically, long travel times to the nearest Community Health Centre and limited cold-chain and referral transport availability during monsoon road closures , rather than by exposure alone. This points to health

system strengthening, rather than disease surveillance alone, as the priority adaptive capacity intervention in these districts (see Section 8).

4.4 Roads and Critical Infrastructure Vulnerability Index

This index, built from the indicator basket detailed in Table 3.1, captures the fragility of Uttarakhand's road and bridge network to landslide and flash-flood disruption, together with the resulting isolation risk for settlements dependent on single-access routes.

Table 4.4: Illustrative district-level Roads and Critical Infrastructure Vulnerability Index scores.

Districts along the Char Dham and high-altitude trekking corridors , Uttarakashi, Chamoli, and Pithoragarh , show the highest exposure sub-scores, reflecting both the geotechnical fragility of the terrain traversed and the seasonal concentration of heavy vehicular and pilgrim traffic that compounds maintenance stress on these routes.

4.5 Tourism and Local Economy Vulnerability Index

Tourism is a significant and growing contributor to district-level economies in Uttarakhand but is highly sensitive to both direct hazard exposure (route closures, infrastructure damage) and seasonal concentration of visitor inflows. This index combines the share of local economic activity dependent on tourism, exposure of tourism infrastructure to hazard zones, and the diversity of the local livelihood base as a proxy for resilience to tourism-sector shocks.

Table 4.5: Illustrative district-level Tourism and Local Economy Vulnerability Index scores.

Pilgrimage-route districts of Uttarakashi, Chamoli, and Rudraprayag , host to the Char Dham circuit , register the highest tourism vulnerability owing to the concentration of livelihoods around a short and weather-sensitive pilgrimage season, combined with direct exposure of trekking routes and roadside tourism infrastructure to landslide and flash-flood hazard.

4.6 Energy Sector Vulnerability Index

Uttarakhand's energy sector vulnerability is driven primarily by the concentration of hydropower generation capacity in flood- and GLOF-exposed river valleys, and by the exposure of transmission infrastructure to landslide-prone corridors. Indicators include the share of installed hydropower capacity within mapped GLOF/flood hazard zones, transmission line length traversing high-landslide-susceptibility terrain, and historical frequency of monsoon-related grid disruption.

Table 4.6: Illustrative district-level Energy Sector Vulnerability Index scores.

Chamoli's elevated energy vulnerability reflects both its high concentration of run-of-river hydropower installations in the upper Alaknanda basin and its history of GLOF and flood-related damage to power infrastructure, underscoring the case for climate risk screening of future hydropower siting decisions in this district.

5. Comprehensive Stakeholder Map & Role of Priority Departments

Effective translation of this Vulnerability Assessment into adaptation action depends on clear identification of the state and district institutions responsible for each sector, and on coordination mechanisms that allow vulnerability and risk information to flow into departmental planning, budgeting, and emergency response processes. This section presents the departmental stakeholder map relevant to the sectors covered in Section 4, together with the interdepartmental coordination architecture required to operationalise the recommendations in Section 8.

List of Departments , Vulnerability and Impact Relevance

Department / Agency	Sectoral Relevance	Role in Vulnerability Reduction
State Environment Conservation & Climate Change Directorate (SECCCD)	Cross-cutting / Climate Policy	Nodal agency for SAPCC implementation, climate vulnerability assessment, and Digital Vulnerability Atlas custodianship
Uttarakhand State Disaster Management Authority (USDMA)	Cross-cutting / Disaster Risk Reduction	Hazard monitoring, early warning dissemination, district disaster management plan coordination
Department of Agriculture & Horticulture	Agriculture & Livelihoods	Crop advisory, climate-resilient seed distribution, crop insurance enrolment, irrigation extension
Irrigation Department / Jal Shakti	Water Resources	Spring-shed management, water harvesting infrastructure, flood control structures
Peyjal Nigam / PHED (Jal Jeevan Mission)	Water Resources / WASH	Drinking water supply infrastructure, source sustainability, water quality monitoring
Public Works Department (PWD)	Roads & Critical Infrastructure	Road design standards, slope stabilisation, landslide-resilient construction, maintenance scheduling
Department of Health & Family Welfare	Human Health	Health facility planning, disease surveillance, emergency referral systems, heat action planning
Forest Department / Uttarakhand Forest Development Corporation	Environment / Forestry	Forest fire management, afforestation, watershed and slope bioengineering
Uttarakhand Power Corporation Limited (UPCL) / UJVNL	Energy	Grid resilience, hydropower risk screening, transmission infrastructure protection
Tourism Development Board	Tourism & Local Economy	Visitor flow management, tourism infrastructure safety standards, livelihood diversification support
Rural Development & Panchayati Raj Department	Social Protection / Livelihoods	MGNREGA convergence for resilience works, livelihood diversification, Self-Help Group support

Department / Agency	Sectoral Relevance	Role in Vulnerability Reduction
District Disaster Management Authorities (DDMAs)	Cross-cutting / Local Response	District-level early warning, evacuation planning, post-disaster damage assessment
Urban Development Department / Municipal Bodies	Urban Infrastructure & Housing	Urban drainage, building bye-law enforcement in hazard zones, urban heat and flood management
Revenue Department	Cross-cutting / Land Records	Land-use records, hazard zone notification, compensation and relief disbursal

Table 5.1: Priority departments and their relevance to climate vulnerability reduction across sectors covered in this assessment.

Interdepartmental Relationships

The table below illustrates the principal coordination linkages required between departments for vulnerability-informed planning to function effectively, organised around three representative high-priority workflows.

Workflow	Lead Department	Key Coordinating Departments	Coordination Requirement
Early warning and evacuation for flash flood / GLOF events	USDMA	DDMAs, Irrigation Dept, PWD, Health Dept	Real-time hazard alert dissemination to district control rooms; pre-positioning of health and PWD response teams
Spring-shed rejuvenation and water security planning	Irrigation Dept / Jal Shakti	Forest Dept, PHED, Rural Development Dept	Joint watershed delineation, afforestation alignment with recharge zones, convergence with MGNREGA works
Climate-resilient road design and post-disaster restoration	PWD	USDMA, Forest Dept, Revenue Dept	Hazard zone data sharing for alignment planning; rapid land clearance and right-of-way coordination post-disaster

Table 5.2: Illustrative interdepartmental coordination requirements for priority vulnerability-reduction workflows.

Interdepartmental Coordination

Three structural gaps in current interdepartmental coordination were identified through this stakeholder mapping exercise and are carried forward as priorities for the adaptation recommendations in Section 8:

- Absence of a shared, block-level vulnerability and hazard data platform accessible to all priority departments, resulting in each department maintaining independent and not always consistent hazard maps; the Digital Vulnerability Atlas under this assignment is intended to directly address this gap.
- Limited routine engagement between the Forest Department and Irrigation Department / Jal Shakti on joint watershed and spring-shed planning, despite the strong technical

interdependence between forest cover and spring discharge sustainability documented in Section 2.b.

- Variable capacity at the District Disaster Management Authority level to translate state-level early warning information into block- and village-level evacuation action, particularly in districts with the highest composite vulnerability scores identified in Section 3.e.

It is recommended that SECCCD convene a periodic (at minimum biannual) interdepartmental coordination forum, drawing on the stakeholder map above, to review updated vulnerability and risk outputs from the Digital Vulnerability Atlas and align departmental annual work plans accordingly. This recommendation is elaborated further within the institutional dimension of the adaptation prioritisation set out in Section 8.

Role of Priority Departments in Vulnerability Reduction

Beyond the coordination workflows summarised in Table 5.2, each priority department identified in Table 5.1 has a distinct and ongoing role in reducing the specific sector vulnerability captured by the corresponding sector index in Section 4. These roles are summarised below to make explicit the link between the stakeholder map and the sector-wise interventions developed in Section 8.

- SECCCD functions as the technical and data custodian for this entire assessment, responsible for maintaining and periodically updating the Composite CVI (Section 6) and the underlying indicator framework (Section 3) as new departmental data becomes available, and for ensuring that the Digital Vulnerability Atlas remains the single authoritative reference for vulnerability and hazard information across departments.
- USDMA and the District Disaster Management Authorities are responsible for translating the block-level risk profiles developed in Section 7 into operational early warning protocols and evacuation plans, with particular priority attached to the high and very high risk blocks identified in Table 7.1.
- Sector line departments (Agriculture & Horticulture, Irrigation/Jal Shakti, PWD, Health, Forest, Energy utilities, and Tourism Development Board) are each responsible for integrating the relevant sector vulnerability index from Section 4 into their annual departmental planning and budgeting cycles, prioritising the high-vulnerability districts and blocks identified within their respective sector tables.
- The Rural Development & Panchayati Raj Department and Revenue Department jointly hold primary responsibility for the social protection and livelihood diversification measures identified in Section 8, given their existing administrative mandate over MGNREGA convergence, Self-Help Group support, and relief and compensation disbursal.

6. Composite Climate Vulnerability Index (CVI)

This section aggregates the sector-specific vulnerability indices presented in Section 4 into a single Composite Climate Vulnerability Index (CVI) for each district, classifies districts into six vulnerability categories, identifies vulnerability hotspots and clusters, and projects how the composite vulnerability profile may shift under future IPCC SSP climate scenarios.

Development of District-Level CVI

The Composite CVI is computed as the equal-weighted average of the eight sector vulnerability indices developed in Section 4 , agriculture and livelihoods (AGVI), water resources (WRVI), human health (CLVI), roads and critical infrastructure, tourism and local economy, energy, social (SVI), and economic (ECVI) , each of which is itself already an aggregation of the IPCC exposure, sensitivity, and adaptive capacity components as set out in Section 3.d. As with the sector indices, a PCA-derived weighting is computed in parallel as a robustness check; where sector weights derived from PCA diverge from equal weighting by more than 0.05 on the normalised 0-1 scale, this is flagged for review during the validation phase.

Because the CVI is constructed entirely from vulnerability-side indicators (exposure, sensitivity, and adaptive capacity, in the IPCC AR5 sense), it does not yet incorporate hazard probability. The CVI presented in this section is therefore a vulnerability index, not a risk index; it answers the question of which districts and population groups are most exposed and least able to cope, independent of how likely a given hazard event is to occur in a given year. The combination of the CVI with hazard probability to generate a risk index, consistent with the IPCC AR6 risk equation, is performed explicitly in Section 7.

Table 6.1: Illustrative district-level Composite Climate Vulnerability Index (CVI), computed as the equal-weighted average of eight sector indices from Section 4.

Vulnerability Classification (VL to EH)

Districts and blocks are classified into six vulnerability categories using a combination of quantile breaks and natural-breaks (Jenks) classification applied to the composite CVI distribution, consistent with the classification approach specified in the Terms of Reference: very low (VL), low (L), moderate (M), high (H), very high (VH), and extremely high (EH). The final thresholds will be re-derived once the complete block-level dataset is available, since natural-breaks classification is sensitive to the full distribution of scores rather than a partial sample.

Category	Composite CVI Range	Illustrative Districts (from Table 6.1)
Extremely High (EH)	0.65 – 1.00	
Very High (VH)	0.56 – 0.64	
High (H)	0.46 – 0.55	
Moderate (M)	0.35 – 0.45	
Low (L)	0.25 – 0.34	
Very Low (VL)	0.00 – 0.24	

Table 6.2: Illustrative vulnerability classification thresholds and district assignment.

Vulnerability Hotspots and Clusters

Cluster analysis of the district-level sector index profiles (Table 6.1) using k-means clustering on the eight standardised sector indices identifies three broad vulnerability clusters across Uttarakhand, which will be refined to the block level once full data are available:

- High Himalayan Multi-Hazard Cluster , Chamoli, Pithoragarh, Uttarkashi, Bageshwar, Rudrapur: characterised by uniformly elevated scores across agriculture, water, roads, and tourism vulnerability, reflecting the compounding effect of terrain fragility, livelihood concentration, and remoteness. This cluster aligns closely with areas of highest landslide and GLOF hazard probability identified in the IIT Roorkee Climate Hazard Profile, and is carried forward as the primary geographic focus of the risk assessment in Section 7.
- Mid-Hill Transition Cluster , Almora, Tehri Garhwal, Pauri Garhwal, Nainital: moderate vulnerability driven predominantly by water resources and agriculture sensitivity, with comparatively better road and health infrastructure access than the high Himalayan cluster, reflecting their intermediate elevation and proximity to district headquarters.
- Plains and Foothills Cluster , Dehradun, Haridwar, Udham Singh Nagar: lowest composite CVI scores, but with localised exception zones along riverine flood corridors and urban drainage-deficient pockets that are masked at the district aggregate level and will require block- or ward-level disaggregation in the final Atlas to be properly represented.

Future Vulnerability Projections

Incorporating SSP Climate Scenarios into Vulnerability

Future vulnerability projections are developed by adjusting the exposure and sensitivity sub-indices computed in Section 3 to reflect projected changes under the IPCC SSP2-4.5 (moderate emissions) and SSP5-8.5 (high emissions) scenarios for the mid-century (2021-2050) and end-century (2071-2100) horizons, using the downscaled climate projections and ETCCDI indices from the IIT Roorkee Climate Hazard Profile. Adaptive capacity indicators are held constant under a business-as-usual institutional scenario, consistent with standard practice in vulnerability projection studies, since adaptive capacity trajectories depend on future policy and investment decisions that are outside the scope of climate model projections; this assumption is made explicit so that projected CVI changes below should be interpreted as a lower bound, achievable adaptive capacity improvements would be expected to moderate the projected increases.

Projected Shifts in Exposure and Sensitivity

Under SSP5-8.5, regional projections indicate a temperature rise of up to 2.3-5.9°C by the end of the century and a 6-16% increase in monsoon precipitation, alongside greater year-to-year variability. Applying these projected shifts to the exposure and sensitivity components of the agriculture, water resources, and roads sector indices indicates a projected increase in composite CVI of approximately 0.08-0.12 (on the 0-1 scale) by mid-century and 0.15-0.22 by end-century for the High Himalayan Multi-Hazard Cluster identified above, under SSP5-8.5, compared to projected increases of roughly half this magnitude under SSP2-4.5.

Future Vulnerability Hotspots

The combination of accelerated glacier melt, increased monsoon precipitation intensity, and projected landslide risk increases of up to 90% in certain districts during monsoon months by the latter half of the century points to an intensification, rather than a spatial shift, of existing vulnerability hotspots: the same high Himalayan districts identified as most vulnerable under current conditions are projected to remain the highest-vulnerability areas under both SSP scenarios, but with a widening gap relative to the mid-hill and plains clusters. This finding directly informs the prioritisation logic applied in Sections 7 and 8.

Note: Quantitative projection ranges above are indicative, derived from applying IIT Roorkee's downscaled SSP projections to the current illustrative indicator sample. These will be recalculated once the finalised, full block-level Climate Hazard Profile dataset and validated vulnerability indicators are integrated.

7. Risk Assessment and Profiling

This section applies the IPCC AR6 risk framework to combine the hazard probability data derived from the IIT Roorkee Climate Hazard Profile with the vulnerability outputs developed in Sections 3, 4, and 6, in order to compute block-level climate risk. Consistent with the AR6 formulation, climate risk in this assessment is treated as arising from the interaction of three terms , hazard, exposure, and vulnerability , rather than as a property of any one term in isolation; a block with very high vulnerability but negligible hazard probability is not classified as high risk under this framework, and conversely a block with high hazard probability but low vulnerability (owing to strong adaptive capacity and limited exposure) is also not classified as high risk.

Combining Hazard Probabilities with Vulnerability Indices

Hazard probability values for each block are drawn directly from the IIT Roorkee Climate Hazard Profile, which provides downscaled, block-disaggregated probability estimates for the principal climate-induced hazards relevant to Uttarakhand , flash floods, landslides, GLOFs, cloudbursts, and forest fires , under both historical and projected (SSP2-4.5 and SSP5-8.5) conditions. These hazard probabilities are normalised to the same 0-1 scale used for the vulnerability indices in Sections 3, 4, and 6, and combined with the Composite CVI using the multiplicative form of the AR6 risk equation:

$$\text{Risk} = f(\text{Hazard probability} \times \text{Exposure} \times \text{Vulnerability})$$

where Vulnerability is the Composite CVI from Section 6 (itself incorporating sensitivity and the inverse of adaptive capacity, consistent with the AR6 sub-components of vulnerability), and Exposure is drawn from the exposure sub-indices computed within each sector index in Section 4, isolated from the sensitivity and adaptive capacity sub-indices for this purpose so as to avoid double-counting exposure within the risk equation. This multiplicative formulation reflects the AR6 understanding that risk requires the joint presence of a hazard, exposed elements, and vulnerable conditions; the absence of any one of the three terms drives realised risk toward zero regardless of the magnitude of the other two.

Block-Level Risk Profiles

Risk Matrix Development

A standard 5×5 risk matrix is constructed by plotting hazard probability (likelihood) against composite vulnerability (a proxy for consequence severity, since higher vulnerability implies greater realised loss and damage for a given hazard magnitude), consistent with conventional disaster risk management practice and with the AR6 risk framework's treatment of risk as the product of likelihood and consequence.

Hazard Probability ↓ / Vulnerability (CVI) →	Very Low	Low	Moderate	High	Very High
Very High	Moderate	High	High	Very High	Very High
High	Low	Moderate	High	High	Very High
Moderate	Low	Low	Moderate	High	High
Low	Low	Low	Low	Moderate	Moderate
Very Low	Low	Low	Low	Low	Moderate

Table 7.2: Illustrative 5×5 risk matrix combining hazard probability and composite vulnerability (CVI) into a qualitative risk classification.

Identification of High-Risk Areas

Applying the risk matrix and the block-level scoring in Table 7.1 to the broader hazard and vulnerability patterns established in Sections 3, 4, and 6, this assessment identifies three priority geographic concentrations of high and very high climate risk that warrant focused adaptation investment, to be confirmed and extended to all blocks once the full dataset is integrated into the Digital Vulnerability Atlas:

- The upper Alaknanda and Mandakini river corridors in Chamoli and Rudrapur districts, where GLOF and flash-flood hazard probability is highest and intersects with high agriculture, tourism, and energy sector vulnerability.
- The Bhagirathi corridor and high-elevation trekking belt of Uttarkashi and Pithoragarh, where landslide hazard probability is highest and intersects with high roads and tourism vulnerability and limited health system adaptive capacity.
- Fire-prone mid-elevation forest belts spanning parts of Almora, Bageshwar, and Nainital, where forest fire hazard frequency (11,256 incidents recorded between November 2023 and June 2024 alone) intersects with high agriculture and water resources sensitivity through downstream impacts on spring recharge and soil stability.

These three priority geographies form the spatial basis for the adaptation and mitigation prioritisation presented in Section 8.

8. Adaptation and Mitigation Prioritisation

This concluding section translates the risk hotspots identified in Section 7 into a prioritised set of adaptation and mitigation measures, organised by sector and by intervention type , nature-based solutions, infrastructure measures, and social protection and livelihood diversification measures , consistent with the IPCC AR6 understanding of adaptation as a dynamic process that directly reduces vulnerability (by strengthening adaptive capacity and reducing sensitivity) and, where feasible, exposure, thereby reducing overall climate risk for a given level of hazard.

Linking Risk Hotspots to Adaptation Needs

Recommendations are anchored explicitly to the three priority risk geographies identified in Section 7: the upper Alaknanda-Mandakini corridor, the Bhagirathi-Pithoragarh trekking belt, and the fire-prone mid-elevation forest belt. For each, the dominant risk driver (whether hazard-, exposure-, or vulnerability-led, per the AR6 decomposition) determines the type of intervention prioritised below , hazard-led risk in GLOF-prone valleys calls primarily for early warning and structural mitigation, whereas vulnerability-led risk in forest-fire-affected agricultural belts calls primarily for sensitivity-reducing and adaptive-capacity-building measures.

Risk Hotspot	Dominant Risk Driver	Priority Intervention Type
Upper Alaknanda-Mandakini corridor (GLOF/flash flood)	Hazard probability (AR6: hazard term)	Early warning systems; structural mitigation (check dams, lake monitoring); hydropower risk screening
Bhagirathi corridor / Pithoragarh trekking belt (landslide)	Exposure + sensitivity (roads, tourism)	Slope bioengineering; resilient road design; tourism season diversification
Mid-elevation forest-fire belt (Almora/Bageshwar/Nainital)	Sensitivity + adaptive capacity (agriculture, water)	Fire management; spring-shed rejuvenation; livelihood diversification

Table 8.1: Linking the three priority risk hotspots from Section 7 to their dominant AR6 risk driver and corresponding intervention focus.

Sector-Wise Priority Interventions